DEPARTMENT OF COMPUTER APPLICATIONS
PSG COLLEGE OF TECHNOLOGY
MCA G1 & G2, SEMESTER 1
26MX16 - PROGRAMMING AND DATA STRUCTURES LABORATORY
PRACTICE SHEET 2

1. A Smart Parking System calculates the parking fee based on the type of vehicle and the number of hours it is parked. The system accepts the vehicle type and parking duration and calculates the total parking fee according to the following rules.

Input Format

The first line contains a string representing the vehicle type:

- BIKE
- CAR
- SUV

The second line contains an integer representing the parking duration in hours.

Constraints

- 1 ≤ hours ≤ 24
- Vehicle type must be BIKE, CAR, or SUV.

Vehicle Type	Rate per Hour
BIKE	₹10
CAR	₹20
SUV	₹30

Output Format

Print the calculated parking fee in the following format:

Parking Fee: ₹<amount>

If the vehicle type is invalid, print:

Invalid Vehicle Type

Sample Input

BIKE
3

Sample Output

Parking Fee: ₹30

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2. A TANCET Admission Management System needs to automatically determine whether a candidate is eligible for admission based on their degree status, TANCET score, and qualifying percentage**.** The system must apply the admission rules and display the candidate's eligibility status.

Input Format

The first line contains a string representing degree status:

- VALID
- INVALID

The second line contains an integer representing the TANCET score.

The third line contains a floating-point number representing the qualifying percentage.

Constraints

- 0 ≤ TANCET Score ≤ 100
- 0 ≤ Qualifying Percentage ≤ 100
- Degree status must be VALID or INVALID.

Business Rules

Rule	Condition
BR1	Degree status is VALID, TANCET score ≥ 50, and qualifying percentage ≥ 50 → Eligible
BR2	TANCET score < 50 → Not Eligible
BR3	Qualifying percentage < 50 → Not Eligible
BR4	Degree status is INVALID → Not Eligible

Output Format

If eligible:

ELIGIBLE

Otherwise:

NOT ELIGIBLE

Sample Input 1

VALID
72
68

Sample Output 1

ELIGIBLE

Sample Input 2

VALID
42
70

Sample Output 2

NOT ELIGIBLE

3. A Ride Booking System calculates the fare for a customer based on the ride type and distance travelled**.**

Input Format

The first line contains a string representing the ride type:
BIKE
AUTO
CAR
The second line contains an integer representing the distance travelled in kilometres.

Constraints

- 1 ≤ distance ≤ 1000
- Ride type must be BIKE, AUTO, or CAR.

Business Rules

Ride Type	Fare per km
BIKE	₹10
AUTO	₹15
CAR	₹20

Output Format

Print:
Fare: ₹<amount>
For an invalid ride type:
Invalid Ride Type

Sample Input 1
BIKE
8
Sample Output 1
Fare: ₹80
Sample Input 2
AUTO
10
Sample Output 2
Fare: ₹150
Sample Input 3
CAR
12
Sample Output 3
Fare: ₹240

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4. A TNEA Admission Management System needs to determine whether a student satisfies the minimum marks specified by the system for engineering counselling. The system receives the student's Mathematics, Physics and Chemistry marks and qualifying examination status.

Input Format

The first three lines contain integer marks for:
Mathematics
Physics
Chemistry
The fourth line contains the qualifying examination status:
PASSED
FAILED

Constraints

- 0 ≤ Mathematics ≤ 100
- 0 ≤ Physics ≤ 100
- 0 ≤ Chemistry ≤ 100

Business Rules

Rule	Condition
BR1	Qualifying examination is PASSED and all three subject marks ≥ 50 → Eligible
BR2	Any subject mark < 50 → Not Eligible
BR3	Qualifying examination is FAILED → Not Eligible

Output Format
If all conditions are satisfied:
ELIGIBLE
Otherwise:
NOT ELIGIBLE

Sample Input 1
85
78
82
PASSED
Sample Output 1
ELIGIBLE
Sample Input 2
85
45
82
PASSED
Sample Output 2
NOT ELIGIBLE
Sample Input 3
90
85
88
FAILED
Sample Output 3
NOT ELIGIBLE

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5. A Municipal Water Management System calculates the monthly water tax based on the consumer type and water consumption.

Input Format
The first line contains the consumer type:
DOMESTIC
COMMERCIAL
The second line contains the monthly water consumption in units.

Constraints

- 0 ≤ consumption ≤ 1000
- Consumer type must be DOMESTIC or COMMERCIAL.

Business Rules

Consumer Type	Consumption	Rate
DOMESTIC	≤ 15 units	₹5/unit
DOMESTIC	> 15 units	₹8/unit
COMMERCIAL	≤ 30 units	₹10/unit
COMMERCIAL	> 30 units	₹15/unit
Any type	0 units	₹50 minimum charge

Output Format
Print:
Water Tax: ₹<amount>

Sample Input 1
DOMESTIC
10
Sample Output 1
Water Tax: ₹50
Sample Input 2
DOMESTIC
25
Sample Output 2
Water Tax: ₹200
Sample Input 3
COMMERCIAL
40
Sample Output 3
Water Tax: ₹600
